Tal Lavian, Ph.D.

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Telecommunications, Computer Networks, Network Communications, and Internet Technologies Expert

Dr. Lavian is a scientist, educator, and technologist with over 35 years of experience. He has co-authored over 25 scientific publications, journal articles, and peer-reviewed papers. He is an expert in network communications and telecommunications, including Internet protocols, data communications, and computer networks.

Dr. Lavian has spent 20 years researching, studying, and lecturing at UC Berkeley. His research and expertise focus on telecommunications, network communications, computer networks, communications systems, network services, network software, network protocols, Internet Protocols TCP/IP, mobile wireless, and communications frameworks.

He holds a Ph.D. in Computer Science from UC Berkeley's College of Engineering, specializing in network communications; an M.Sc. in Electrical Engineering from Tel Aviv University; and a B.Sc. in Mathematics and Computer Science.

EXPERTISE

Network communications, telecommunications, Internet protocols, and mobile wireless:

• **Network Communications:** Internet protocols; TCP/IP suite, TCP, UDP, IP, Ethernet, 802.3, LAG, network protocols, network software applications, data link, network, transport layers, SNMP, NMS, network management, packet switching, and network architecture.

• **VoIP/Streaming Media:** VoIP, SIP, RTP, video/audio conferencing, streaming media, IP telephony, transport systems, PSTN, circuit switching, SS7, SONET, SDH, and TDM.

• **Mobile wireless:** Wi-Fi, 802.11, Bluetooth, 802.15, Wireless LAN (WLAN), MAC, PHY, ARQ, HARQ. Cellular, SMS, MMS, messaging, chat, mobile devices, and smartphones.

• **Internet/cloud:** Internet Technologies, Web applications, HTTP, HTTPS, cybersecurity, firewalls, SSH, SSL, FTP, e-mail, client-server, cloud computing, and distributed computing.

• **Routing/switching:** LAN, WAN, VPN, encapsulation, routing protocols, RIP, BGP, MPLS, OSPF, multicast, VPLS, Pseudowire, DNS, QoS, queuing, traffic control, network security, deep packet inspection, L4-L7 switching, servers, gateways, computer networks, data communications, communications systems, network infrastructure, and architectures.

Dr. Lavian has extensive experience in software development for computer networks, architectures, configurations, installations, and network testing. He has academic and hands-on experience in the above fields, including technology products from various companies, implementations, related standards, designs, systems, hardware, and software.

ACCOMPLISHMENTS

- Principal Investigator (PI) for three US Department of Defense (DARPA) projects.
- Directed networking computation project for the US Air Force Research Lab (AFRL). PI of a wireless research project for an undisclosed US federal agency.
- An inventor of over 120 patents, over 60 prosecuted *pro-se* before the USPTO.
- Led and developed the first network resource scheduling service for grid computing.

- Managed and engineered the first demonstrated dynamic transatlantic allocation of 10 Gbs Lambdas as a grid service.
- Developed and successfully demonstrated the first wire-speed active network on commercial hardware.
- Created and chaired Nortel Networks' EDN Patent Committee.

EXPERT WITNESS

Dr. Lavian has served as an expert witness in cases in over 90 cases, providing expert reports and testimony in over 60 depositions. He has testified in arbitrations, State and Federal courts before judges and juries, the USPTO PTAB, the ITC, and International Courts. These cases involved leading companies such as Google, Amazon, LinkedIn, Avaya, Netflix, T-Mobile, ZTE, Ericsson, Cisco Systems, Juniper Networks, Arista Networks, Polycom, Motorola, LG, WhatsApp, Instagram, Delta Airlines, Microsoft, Huawei, Facebook, and Apple.

PROFESSIONAL EXPERIENCE

The University of California, Berkeley, Berkeley, California 2000-2019
U.C. Berkeley SkyDeck, Industry Fellow, Lecturer, Visiting Scientist, Ph.D. Candidate, Nortel's Scientist Liaison
Some positions and projects were concurrent, others sequential.

- U.C. Berkeley SkyDeck startups - advanced technology research, development, business, and market.
- Industry fellow and lecturer at the Sutardja Center for Entrepreneurship and Technology (SCET).
- Conducted research projects in data centers (RAD Labs), telecommunication infrastructure (SAHARA), and wireless systems (ICEBERG).
- Acted as a scientific liaison between Nortel Research Lab and U.C. Berkeley, providing tangible value in advanced technologies.
- Developed long-term technology for the enterprise market, integrating communication and computing technologies.
- Studied network services, telecommunication systems and software, communications infrastructure, and data centers.
- Earned a Ph.D. in Computer Science with a specialization in network communications.

TelecommNet Engineering, Inc., Encino, California 2006-Present
Principal Scientist

- Consulting on system architecture and technology analysis for telecommunications, computer networks, data communication, network communications, mobile wireless, network protocols, Internet Protocols (TCP/IP), and Internet web technology projects.
- Consulting and expert witness practice serving leading IP litigation firms and technology companies in matters involving telecommunications, network communications, and emerging communications technologies and architectures.

- Expert reports on infringement, invalidity, non-infringement, claim construction (Markman), and rebuttal opinions across patent litigation, telecommunications, and commercial disputes.
- Providing expert witness services in telecommunications, computer networks, and network communications patent infringement lawsuits, including invalidity and infringement claim charts.
- Prior art investigations, source code review, claim charts, and standards-essential patent (SEP) analysis across IEEE, IETF, 3GPP, and ITU standards.
- Forensic analyses of telephony Call Detail Records (CDRs), VoIP traffic, email headers and routing, cellular tower data, IP geolocation, and telecommunications infrastructure failures for civil and criminal matters.
- Management and marketing under TIL Manage-Marketing.
- System architecture, technology assessments, and standards analysis for computer networks, mobile wireless devices, IoT systems, and Internet web technologies.
- Independent technology assessments and prior art investigations for product strategy, due diligence, and investment decisions.

CRadar.Ai, U.C. Berkeley, California 2018-2019
CTO / Principal Investigator
- CRadar.Ai improves the Radar wireless RF signal phase noise purity by 100x.
- Accurate Radars are paramount for self-driving car safety. Radars “see” where Cameras and LiDars are “blind” (fog, rain, snow, direct sunlight, and darkness).
- The superior wireless RF signal quality provides a clean signal for high Radar accuracy.
- Improving Radar accuracy and resolution enables genuine redundancy and sensory fusion and puts the Radar into the sensory spearhead.

Aybell (VisuMenu Inc.), U.C. Berkeley, California 2016-Present
CEO/CTO
- Aybell transforms smartphones into visual menu systems, making the phone a frictionless point for user interactions with customer service platform features. Empower consumers to reach suitable agents in call centers, overcoming customer service barriers. Aybell is a brand and marketer of VisuMenu advanced technologies.
- Architecture, design, and implementation of a cloud data center for connecting smartphone users to any company and service by digitizing interactive voice systems and exposing APIs to other applications through cloud service.
- The system was deployed as a cloud networking and cloud computing service on Amazon Web Services (AWS) and Google Cloud Platform (GCP).
- Technologies include Data Science analytics, Machine Learning (ML), Artificial Intelligence (AI), and Statistical Learning (SL). Building an NLP Parser using Python and other NLP libraries and modules.

VisuMenu, Inc., Sunnyvale, California 2010-2016
Co-Founder and Chief Technology Officer (CTO)

- Led the software design and development of a visual IVR system for smartphones and other mobile devices based on the innovative use of wireless and network communications technologies.
- Designed a voice search engine for IVR / PBX using Asterisk, SIP, and VoIP.
- The system was deployed as a cloud networking and cloud computing service on Amazon Web Services (AWS) and Google Cloud Platform (GCP).
- VisuMenu advanced technologies rebranded as Aybell.

Ixia, Santa Clara, California 2008 - 2008

Network Communications Consultant

Researched and developed advanced network communications testing technologies:

- IxNetwork/IxN2X —IP routing, switching devices, and broadband access equipment. Provided traffic generation and emulation for the full range of protocols: OSPF, RIP, EIGRP, BGP, IS-IS, MPLS, unicast, multicast, broadcast, layer 2/3 VPNs, IPSec, carrier Ethernet, broadband access, and data center bridging. Tested and validated compatibility with IEEE, ITU, and IETF RFC standards.
- IxLoad — quickly and accurately modeled high-volume video, data, and voice subscribers and servers to test the real-world performance of multiservice delivery and security platforms.
- IxCatapult emulated a broad range of wireless access and core protocols to test wireless components and systems. When combined with IxLoad, this provides an end-to-end solution for testing wireless service quality.
- IxVeriWave — employed a client-centric model to test Wi-Fi and wireless LAN networks by generating repeatable, large-scale, real-world test scenarios that are virtually impossible to create by any other means.
- Test automation — provided simple, comprehensive lab automation to help test engineering teams create, organize, catalog, and schedule the execution of tests.

Nortel Networks, Santa Clara, California, 1996 - 2007

Employed initially by Bay Networks, later acquired by Nortel Networks

Principal Scientist, Principal Architect, Principal Engineer, Senior Software Engineer

Held scientific and research roles at Nortel Labs, Bay Architecture Labs, and the CTO's office.

Principal Investigator for U.S. Department of Defense (DARPA) Projects

- Conceived, proposed, and completed three research projects: active networks, DWDM-RAM, and a networking computation project for the Air Force Research Lab (AFRL).
- Led a wireless research project for an undisclosed U.S. federal agency.

Academic and Industrial Researcher

- Analyzed new technologies to reduce risks associated with R&D investment.
- Led research collaborations with leading universities and professors at UC Berkeley, Northwestern University, the University of Amsterdam, and the University of Technology, Sydney.
- Evaluated competitive products relative to Nortel's products and technology.

- Proactively identified prospective business ideas, leading to new networking products.
- Predicted technological trends through researching the technological horizon and academic sphere.
- Designed software for switches, routers, and network communications devices.
- Developed systems and architectures for switches, routers, and network management.
- Researched and developed the following projects:
  - Data-Center Communications: network and server orchestration 2006-2007
  - DRAC: SOA-facilitated L1/L2/L3 network dynamic controller 2003-2007
  - Omega: classified project for undisclosed U.S. Federal Agency 2006-2006
  - Platform project for the U.S. Air Force Research Laboratory (AFRL) 2005-2005
  - Network resource orchestration for Web services workflows 2004-2005
  - A proxy study between Web/grids services and network services 2004-2004
  - Streaming content replication: real-time A/V media multicast at edge 2003-2004
  - DWDM-RAM: U.S. DARPA-funded program on agile optical transport 2003-2004
  - Packet capturing and forwarding service on IP and Ethernet traffic 2002-2003
  - CO2: content-aware agile networking 2001-2003
  - Active networks: US DARPA-funded research program 1999-2002
  - ORE: programmable network service platform 1998-2002
  - JVM platform: Java on network devices 1998-2001
  - Web-based device management: network device management 1996-1997

Technology Innovator and Patent Leader

- Created and chaired Nortel Networks' EDN Patent Committee.
- Facilitated a continuous stream of innovative ideas and their conversion into intellectual property rights.
- Developed intellectual property assets through invention and analysis of existing technology portfolios.

Aptel Communications, Netanya, Israel 1994-1995

Software Engineer, Team Leader

Start-up company focused on mobile wireless CDMA spread spectrum PCN/PCS.

- Developed a mobile wireless device using an unlicensed band - Direct Sequence Spread Spectrum (DSSS); FCC part 15 - unlicensed transmitters.
- Designed and managed a personal communication network (PCN) and personal communication system (PCS), which were the precursors of short text messages (SMS).
- Designed and developed network communications software products in C/C++.
- Invented and implemented a two-way paging product.

Scitex Ltd., Herzeliya, Israel 1990-1993

Software Engineer, Team Leader

Software and hardware company acquired by Hewlett-Packard (HP)

- Developed system and network communications in C/C++.
- I provided IT services, System Administration, and network administration.
- I worked on Unix systems, including IBM AIX, HP, and SUN Unix.

- Invented Parallel SIMD Architecture.
- Participated in the Technology Innovation group.

Shalev, Ramat-HaSharon, Israel 1987-1990
Start-up company
Software Engineer
- Developed real-time software and algorithms in C/C++ and Pascal.

PROFESSIONAL ASSOCIATIONS

- IEEE senior member
- IEEE CNSV co-chair, Intellectual Property SIG (2013)
- President Next Step Toastmasters (an advanced TM club in the Silicon Valley) (2013-2014)
- Technical co-chair, IEEE Hot Interconnects 2005 at Stanford University
- Member, IEEE Communications Society (COMMSOC)
- Member, IEEE Computer Society
- Member, IEEE Systems, Man, and Cybernetics Society
- Member, IEEE-USA Intellectual Property Committee (2012)
- Member, ACM, ACM Special Interest Group on Data Communication (SIGCOM)
- Member, ACM Special Interest Group on Hypertext, Hypermedia, and Web (SIGWEB)
- Member, IEEE Consultants' Network (CNSV)
- Global Member, Internet Society (ISOC)
- President Java Users Group – Silicon Valley Mountain View, CA,1999-2000
- Toastmasters International

FORMER ADVISORY BOARD POSITIONS

- Quixey –search engine for wireless mobile apps
- Mytopia – mobile wireless social games
- iLeverage – Israeli Innovations

PROFESSIONAL AWARDS

- Top Talent Award – Nortel
- Top Inventors Award – Nortel EDN
- Certified IEEE-WCET - Wireless Communications Engineering Technologies (2012)
- Toastmasters International - Competent Communicator (twice)
- Toastmasters International - Advanced Communicator Bronze
- Best Paper Presentation Award - ICE/IEEE Conference. "R&D Models for Advanced Development & Corporate Research"

PERSONAL

- USA FIT – San Jose Marathon running club (2017-2020)

Patents and Publications

Patents Issued
(Representative List)

US 9,690,877	Systems and methods for electronic communications	Link
US 9,660,655	Ultra-low phase noise frequency synthesizer	Link
US 9,184,989	Grid proxy architecture for network resources	Link
US 9,521,255	Systems and methods for visual presentation and selection of IVR menu	Link
US 9,083,728	Systems and methods to support sharing and exchanging in a network	Link
US 9,021,130	Photonic line sharing for high-speed routers	Link
US 8,762,963	Translation of programming code	Link
US 8,762,962	Methods and apparatus for automatic translation of a computer program language code	Link
US 8,745,573	Platform-independent application development framework	Link
US 8,731,148	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,688,796	Rating system for determining whether to accept or reject an objection raised by a user in a social network	Link
US 8,619,793	Dynamic assignment of traffic classes to a priority queue in a packet-forwarding device	Link
US 8,572,303	A portable Universal communication device	Link
US 8,553,859	Device and method for providing enhanced telephony	Link
US 8,548,131	Systems and methods for communicating with an interactive voice response system	Link
US 8,537,989	Device and method for providing enhanced telephony	Link
US 8,341,257	Grid proxy architecture for network resources	Link
US 8,161,139	Method and apparatus for intelligent management of a network element	Link
US 8,146,090	Time-value curves to provide dynamic QoS for time-sensitive file transfer	Link
US 8,078,708	Grid proxy architecture for network resources	Link

US 7,944,827	Content-aware dynamic network resource allocation	Link
US 7,860,999	Distributed computation in network devices	Link
US 7,734,748	Method and apparatus for intelligent management of a network element	Link
US 7,710,871	Dynamic assignment of traffic classes to a priority queue in a packet-forwarding device	Link
US 7,580,349	Content-aware dynamic network resource allocation	Link
US 7,433,941	Method and apparatus for accessing network information on a network device	Link
US 7,359,993	Method and apparatus for external interfacing resources with a network element	Link
US 7,313,608	Method and apparatus for using documents written in a markup language to access and configure network elements	Link
US 7,260,621	The object-oriented network management interface	Link
US 7,237,012	Method and apparatus for classifying Java remote method invocation transport traffic	Link
US 7,127,526	Method and apparatus for dynamically loading and managing software services on a network device	Link
US 7,047,536	Method and apparatus for classifying remote procedure call transport traffic	Link
US 7,039,724	Programmable command-line interface API for managing the operation of a network device	Link
US 6,976,054	Method and system for accessing low-level resources in a network device	Link
US 6,970,943	Routing architecture includes a compute plane configured for high-speed processing of packets to provide application layer support.	Link
US 6,950,932	Security association mediator for Java-enabled devices	Link
US 6,850,989	Method and apparatus for automatically configuring a network switch	Link
US 6,845,397	Interface method and system for accessing inner layers of a network protocol	Link
US 6,842,781	Download and processing of a network management application on a network device	Link
US 6,772,205	Executing applications on a target network device using a proxy network device	Link
US 6,564,325	Method of and apparatus for providing multi-level security access to a system	Link
US 6,175,868	Method and apparatus for automatically configuring a network switch	Link
US 6,170,015	Network apparatus with Java co-processor	Link

US 8,687,777	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,681,951	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,625,756	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,594,280	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,548,135	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,406,388	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,345,835	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,223,931	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,160,215	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,155,280	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,054,952	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,000,454	Systems and methods for visual presentation and selection of IVR menu	Link
EP 1,905,211	A technique for authenticating network users	Link
EP 1,142,213	Dynamic assignment of traffic classes to a priority queue in a packet forwarding device	Link
EP 1,671,460	Method and apparatus for scheduling resources on a switched underlay network	Link
US 9,001,819	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,949,846	Time-value curves to provide dynamic QoS for time-sensitive file transfers	Link
US 8,929,517	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,903,073	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,898,274	Grid proxy architecture for network resources	Link
US 8,880,120	Device and method for providing enhanced telephony	Link
US 8,879,703	System method and device for providing tailored services when a call is on-hold	Link
US 8,879,698	Device and method for providing enhanced telephony	Link
US 8,867,708	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,787,536	Systems and methods for communicating with an interactive voice response system	Link
US 8,782,230	Method and apparatus for using a command design pattern to access and	Link

	configure network elements	
CA 2,358,525	Dynamic assignment of traffic classes to a priority queue in a packet-forwarding device	Link
CA 2,989,752	Ultra-low Phase Noise Frequency Synthesizer	Link
US 10,598,764	Radar target detection and imaging system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 10,404,261	Radar target detection system for autonomous vehicles with an ultra-low phase-noise frequency synthesizer	Link
US 10,348,313	Radar target detection system for autonomous vehicles with an ultra-low phase-noise frequency synthesizer	Link
US 10,205,457	RADAR target detection system for autonomous vehicles with an ultra-low phase-noise frequency synthesizer	Link
US 10,764,264	Technique for authenticating network users	Link
EP 3,311,493	An ultra-low phase-noise frequency synthesizer	Link
US 9,831,881	Radar target detection system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 9,762,251	Ultra-low phase noise frequency synthesizer	Link
US 9,729,158	Ultra-low phase noise frequency synthesizer	
US 9,705,511	Ultra-low phase noise frequency synthesizer	Link
WO 2000/041368	Dynamic assignment of traffic classes to a priority queue	
WO 2005/033899	Method and apparatus for scheduling resources on a switched underlay network	
AU 22249/00	Dynamic assignment of traffic classes to a priority queue	
AU 37405/00	Method and apparatus for accessing network information on a network device	
DE 600 24 228 T2	Dynamic assignment of traffic classes to a priority queue	

Patent Applications and Published
(Representative List)

US 20150058490	Grid Proxy Architecture for Network Resources	Link
US 20150010136	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20140379784	Method and Apparatus for Using a Command Design Pattern to Access and Configure Network Elements	Link
US 20140105025	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
US 20140105012	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
US 20140012991	Grid Proxy Architecture for Network Resources	Link
US 20130080898	Systems and Methods for Electronic Communications	Link
US 20130022191	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20130022183	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20130022181	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20120180059	Time-Value Curves to Provide Dynamic QoS for Time Sensitive File Transfers	Link
US 20120063574	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20110225330	Portable Universal Communication Device	Link
US 20100220616	Optimizing Network Connections	Link
US 20100217854	Method and Apparatus for Intelligent Management of a Network Element	Link
US 20100146492	Translation of Programming Code	Link
US 20100146112	Efficient Communication Techniques	Link
US 20100146111	Efficient Communication in a Network	Link
US 20090313613	Methods and Apparatus for Automatic Translation of a Computer Program Language Code	Link
US 20090313004	Platform-Independent Application Development Framework	Link
US 20090279562	Content-aware dynamic network resource allocation	Link
US 20080040630	Time-Value Curves to Provide Dynamic QoS for Time Sensitive File	Link

	Transfers	
US 20070169171	A technique for authenticating network users	Link
US 20060123481	Method and apparatus for network immunization	Link
US 20060075042	Extensible Resource Messaging Between User Applications and Network Elements in a Communication Network	Link
US 20050083960	Method and Apparatus for Transporting Parcels of Data Using Network Elements with Network Element Storage	Link
US 20050076339	Method and Apparatus for Automated Negotiation for Resources on a Switched Underlay Network	Link
US 20050076336	Method and Apparatus for Scheduling Resources on a Switched Underlay Network	Link
US 20050076173	Method And Apparatus for Preconditioning Data to Be Transferred on a Switched Underlay Network	Link
US 20050076099	Method and Apparatus for Live Streaming Media Replication in a Communication Network	Link
US 20050074529	Method and apparatus for transporting visualization information on a switched underlay network	Link
US 20040076161	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
US 20020021701	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
WO 2006/063052	Method and apparatus for network immunization	Link
WO 2007/008976	A technique for authenticating network users	Link
WO2000/054460	Method and apparatus for accessing network information on a network device	Link
WO/2016/203460	Ultra-low phase noise frequency synthesizer	Link
WO/2005/033899	Method and apparatus for scheduling resources on a switched underlay network	Link
WO/2000/041368	Dynamic assignment of traffic classes to a priority queue in a packet forwarding device	Link
US 20140156556	A Time-variant rating system and method thereof	Link
US 20140156758	A Reliable rating system and method thereof	Link

US 20170085708	Systems and methods for visual presentation and selection of IVR menu	Link
US 20160373117	Ultra-low phase noise frequency synthesizer	Link
US 20170322687	Systems and methods for electronic communications	Link
US 20170302282	Radar target detection system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 20180019755	Radar target detection system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 20170289332	Systems and methods for visual presentation and selection of IVR menu	Link
US 20170269797	Systems and methods for electronic communication	Link
US 20170099058	Ultra-low phase noise frequency synthesizer	Link
US 20170099057	Ultra-low phase noise frequency synthesizer	Link
US 20220043108	Systems, Methods, and Apparatus for Deep-Learning Multidimensional Detection, Segmentation, and Classification	
US 20210208272	Radar Target Detection System for Autonomous Vehicles with Ultra-Low Phase-Noise Frequency Synthesizer	
US 20190128998	Radar target detection and imaging system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 20190082043	Systems and methods for visual presentation and selection of IVR menu	Link
US 20180146090	Systems and methods for visual presentation and selection of IVR menu	Link
US 20180130102	Reliable rating system and method thereof	Link
WO 2016/203460	Ultra-low phase noise frequency synthesizer	
AU 2016/279027	Ultra-low phase noise frequency synthesizer	
KR 20180095793	Ultra-low phase noise frequency synthesizer	
IL 256,357	Ultra-low phase noise frequency synthesizer	
RU 2018/101470	Ultra-low phase noise frequency synthesizer	
JP 2019-504520 A	Ultra-low phase noise frequency synthesizer	

[CN 107925413 A]()	[Ultra-low phase noise frequency synthesizer]()
[SG 11201710413 S]()	[Ultra-low phase noise frequency synthesizer]()
[MY PI2017704822]()	[Ultra-low phase noise frequency synthesizer]()

Publications
(Representative List)

- "R&D Models for Advanced Development & Corporate Research" Understanding Six Models of Advanced R&D - Ikhlaq Sidhu, Tal Lavian, Victoria Howell - University of California, Berkeley. ASEE Annual Conference and Exposition- 2015. Received "Best Paper Presentation Award" ICE/IEEE Conference June 2015.
- "Communications Architecture in Support of Grid Computing," Tal Lavian, Scholar's Press 2013ISBN 978-3-639-51098-0.
- "Applications Drive Secure Light-path Creation across Heterogeneous Domains, Feature Topic Optical Control Planes for Grid Networks: Opportunities, Challenges, and the Vision." Gommans L.; Van Oudenaarde B.; Dijkstra F.; De Laat C.; Lavian T.; Monga I.; Taal A.; Travostino F.; Wan A.; IEEE Communications Magazine, vol. 44, no. 3, March 2006, pp. 100-106.
- *Lambda Data Grid: Communications Architecture in Support of Grid Computing*. Tal I. Lavian, Randy H. Katz; Doctoral Thesis, University of California at Berkeley. January 2006.
- "Information Switching Networks." Hoang D.B.; T. Lavian; The 4th Workshop on the Internet, *Telecommunications and Signal Processing, WITSP2005*, December 19-21, 2005, Sunshine Coast, Australia.
- "Impact of Grid Computing on Network Operators and HW Vendors." Allcock B.; Arnaud B.; Lavian T.; Papadopoulos P.B.; Hasan M.Z.; Kaplow W.; *IEEE Hot Interconnects at Stanford University 2005*, pp.89-90.
- *DWDM-RAM: A Data Intensive Grid Service Architecture Enabled by Dynamic Optical Networks*. Lavian T.; Mambretti J.; Cutrell D.; Cohen H.J; Merrill S.; Durairaj R.; Daspit P.; Monga I.; Naiksatam S.; Figueira S.; Gutierrez D.; Hoang D.B., Travostino F.; *CCGRID 2004*, pp. 762-764.
- *DWDM-RAM: An Architecture for Data-Intensive Service Enabled by Next Generation Dynamic Optical Networks*. Hoang D.B.; Cohen H.; Cutrell D.; Figueira S.; Lavian T.; Mambretti J.; Monga I.; Naiksatam S.; Travostino F.; Proceedings IEEE Globecom 2004, Workshop on High-Performance Global Grid Networks, Houston, 29 Nov. to 3 Dec. 2004, pp.400-409.
- *Implementation of a Quality of Service Feedback Control Loop on Programmable Routers*. Nguyen C.; Hoang D.B.; Zhao, I.L.; Lavian, T.; Proceedings, 12th IEEE International Conference on Networks 2004. (ICON 2004) Singapore, Volume 2, 16-19 Nov. 2004, pp.578-582.
- *A Platform for Large-Scale Grid Data Service on Dynamic High-Performance Networks*. Lavian T.; Hoang D.B.; Mambretti J.; Figueira S.; Naiksatam S.; Kaushil N.; Monga I.; Durairaj R.; Cutrell D.; Merrill S.; Cohen H.; Daspit P.; Travostino F; GridNets 2004, San Jose, CA., October 2004.
- *DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks*. Figueira S.; Naiksatam S.; Cohen H.; Cutrell D.; Daspit, P.; Gutierrez D.; Hoang D. B.; Lavian T.; Mambretti J.; Merrill S.; Travostino F; Proceedings, 4th IEEE/ACM International Symposium on Cluster Computing and the Grid, Chicago, USA, April 2004, pp. 707-714.

- *DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks.* Figueira S.; Naiksatam S.; Cohen H.; Cutrell D.; Gutierrez D.; Hoang D.B.; Lavian T.; Mambretti J.; Merrill S.; Travostino F.; 4th IEEE/ACM International Symposium on Cluster Computing and the Grid, Chicago, USA, April 2004.

- *An Extensible, Programmable, Commercial-Grade Platform for Internet Service Architecture.* Lavian T.; Hoang D.B.; Travostino F.; Wang P.Y.; Subramanian S.; Monga I.; IEEE Transactions on Systems, Man, and Cybernetics on Technologies Promoting Computational Intelligence, Openness and Programmability in Networks and Internet Services Volume 34, Issue 1, Feb. 2004, pp.58-68.
- *DWDM-RAM: An Architecture for Data-Intensive Service Enabled by Next Generation Dynamic Optical Networks*. Lavian T.; Cutrell D.; Mambretti J.; Weinberger J.; Gutierrez D.; Naiksatam S.; Figueira S.; Hoang D. B.; Supercomputing Conference, SC2003 Igniting Innovation, Phoenix, November 2003.
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Presentations and Talks
(Not an exhaustive list)

- Lambda Data Grid
- A Platform for Large-Scale Grid Data Service on Dynamic High-Performance Networks
- Lambda Data Grid: An Agile Optical Platform for Grid Computing and Data-intensive Applications.
- Workflow Integrated Network Resource Orchestration
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- Impact of Grid Computing on Network Operators and HW Vendors
- Web Services and OGSA
- WINER Workflow Integrated Network Resource Orchestration.
- A Grid Proxy Architecture for Network Resources
- Technology & Society
- Abundant Bandwidth and how it affects us?
- Active Content Networking (ACN)
- DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks
- Application-engaged Dynamic Orchestration of Optical Network Resources
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- An Architecture for Data-Intensive Service Enabled by Next Generation Optical Networks
- A Platform for Data-Intensive Services Enabled by Next Generation Dynamic Optical Networks
- A Platform for Data-Intensive Services Enabled by Next Generation Dynamic Optical Networks
- Optical Networks
- Grid Optical Network Service Architecture for Data-Intensive Applications
- Optical Networking & DWDM
- OptiCal Inc.
- OptiCal & LUMOS Networks
- Optical Networking Services
- Optical Networks
- Business Models for Dynamically Provisioned Optical Networks
- Business Model Concepts for Dynamically Provisioned Optical Networks
- Optical Networks Infrastructure
- Research Challenges in Agile Optical Networks
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- Impact on Society
- Technology & Society
- TeraGrid Communication and Computation
- Unified Device Management via Java-enabled Network Devices
- Active Network Node in Silicon-Based L3 Gigabit Routing Switch
- Enabling Active Flow Manipulation (AFM) in Silicon-based Network Forwarding Engines
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- Active Nets Technology Transfer through High-Performance Network Devices
- Enabling Active Network Services on A Gigabit Routing Switch
- Programmable Network Node: Applications
- Open Innovation via Java-enabled Network Devices
- Practical Considerations for Deploying a Java Active Networking Platform
- Open Programmable Architecture for Java-enabled Network Devices
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- Open Programmable Architecture for Java-enabled Network Devices
- Open Java-based Intelligent Agent Architecture for Adaptive Networking Devices
- Edge Device Multi-unicasting for Video Streaming
- Intelligent Network Services through Active Flow Manipulation
- Java SNMP Oplet
- Unified Device Management via Java-enabled Network Devices
- Dynamic Classification in a Silicon-Based Forwarding Engine
- Integrating Active Networking and Commercial-Grade Routing Platforms
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- Open Distributed Networking Intelligence: A New Java Paradigm
- Open Networking: Better Networking Through Programmability
- Open Programmability
- Active On A Programmable Networking Platform
- Open Networking through Programmability
- Open Programmable Architecture for Java-enabled Network Devices
- Popeye – Fine-grained Network Access Control for Mobile Users
- Integrating Active Networking and Commercial-Grade Routing Platforms
- Active Networking
- Programmable Network Devices
- Open Programmable Architecture for Java-enabled Network Devices
- To be smart or not to be?